

Local optima

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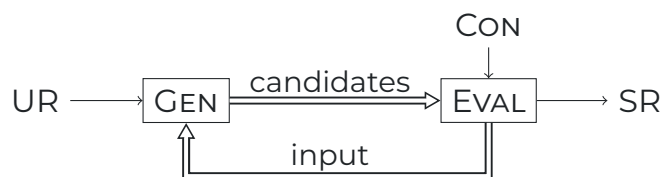
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[Harmonic Serialism](#)¹ (Prince and Smolensky, 1993/2004; McCarthy, 2000) – or more aptly *persistent OT* (McCarthy, 2006, 2007)

¹ [Harmonic Serialism](#)

- CON: ranked phonotactic and faithfulness constraints
no [Correspondence Theory](#)² (McCarthy and Prince, 1994, 1995, 1999)
- GEN: returns candidates that differ from input with at most one operation application
requires explicit hypotheses about operations
- Generate and evaluate until input is optimal

² [Correspondence Theory](#)



Today: evidence for HS from non-optimal mappings

1 Nasal prefixes in Bukusu ([bxk](#), Niger-Congo)

Three processes at prefix-root boundary; let's ignore the behavior of the infinitive and the tonal alternations

- Post-nasal voicing
/N-[t](#)ím-a/ → [n-[d](#)ím-a]
- Post-nasal hardening
/N-[β](#)ukul-a/ → [m-[b](#)ukúl-a]
- Nasal deletion
/N-[sáβ](#)-a/ → [sáβ-a]

(1) **Segmental alternations in Bukusu** (Mutonyi, 2000:168, 173-175)

	/xu-xu-/	/N-/	
a. /-tim-/	xu:-tim-a	n-dím-a	'to run / I run'
b. /-kaβ-/	xu:-kaβ-a	ŋ-gáβ-a	'to divide / I divide'
c. /-βukul-/	xú:-βukúl-á	m-bukúl-a	'to take / I take'
d. /-rafu-/	xú:-rafú-á	n-dafú-a	'to get angry / I get angry'
e. /-xalakil-/	xú:-xalákíl-á	xalákil-a	'to cut fast; to decide / I cut fast; I decide'
f. /-saβ-/	xu:-saβ-a	sáβ-a	'to beg / I beg'

1.1 Parallel OT does not work

These alternations don't work in OT with these constraints

- $*[+nasal][-voice]$: Assign One Violation For Every nasal followed by a voiceless obstruent.³
- $*[+nasal][+cont]$: AOVFE nasal followed by a continuant obstruent.⁴
- IDENT(nasal): AOVFE changed [nasal] feature.⁵
- IDENT(voice): AOVFE changed [voice] feature.
- IDENT(cont): AOVFE changed [continuant] feature.
- DEP: AOVFE inserted segment.⁶
- MAX: AOVFE deleted segment.⁷
- MAX/root: AOVFE deleted root segment.

³ Pater (1999:313)

⁴ Zsiga et al. (2006:725)

⁵ McCarthy and Prince (1994:340)

⁶ McCarthy and Prince (1994:342)

⁷ McCarthy and Prince (1994:116, 342)

Changing [voice] (e) is preferred to changing [nasal] (b), epenthesis (c), and deletion (d)⁸

(2) /N-tím-a/ → [n-dím-a]

/N-tím-a/	$*[+nasal][-voice]$	$*[+nasal][+cont]$	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [ntíma]	W 1					L		
b. [dtíma]			W 1			L		
c. [nətíma]				W 1		L		
d. [tíma]					W 1	L		
 e. [ndíma]						1		

⁸ W = more violations than winner; L = fewer violations than winner; rankings are valid when every L is dominated by some W (Prince, 2002)

Changing [continuant] (e) is preferred to changing [nasal] (b), epenthesis (c), and deletion (d)

(3) /N-βukul-a/ → [m-bukúl-a]

/N-βukul-a/	*[+nasal][-voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [mβukúla]		W 1					L	
b. [bβukúla]			W 1				L	
c. [nəβukúla]				W 1			L	
d. [βukúla]					W 1		L	
 e. [mbukúla]							1	

Deletion (d) is preferred to changing [voice] and [cont] (f)

(4) /N-sáβ-a/ → [sáβ-a]

/N-sáβ-a/	*[+nasal][-voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [nsáβa]	W 1	W 1			L			
b. [dsáβa]			W 1		L			
c. [nəsáβa]				W 1	L			
 d. [sáβa]					1			
e. [náβa]					1			W 1
 f. [ndáβa]					L	W 1	W 1	

Parallel OT does not predict this

- Deletion is more expensive than changing features

$$\langle 0, 0, 0, 0, 1, 0, 0, 0 \rangle > \langle 0, 0, 0, 0, 0, m, n, 0 \rangle \quad \forall m, n$$

- Changing both features is the global optimum;⁹ its violation vector¹⁰ is the lowest

$$\langle 0, 0, 0, 0, 0, 1, 1, 0 \rangle < \langle 0, 0, 0, 0, 1, 0, 0, 0 \rangle$$

- OT only returns global optima

“for a given input i, a grammatical structural description or output A is one that has the

⁹ global optimum

¹⁰ violation vector

greatest Harmony among the set $\text{Gen}(i)$ of all possible parses of i . That is, a parse A of i is grammatical if and only if there is no competing parse B in $\text{Gen}(i)$ such that B has higher Harmony than A " (Prince and Smolensky, 1993/2004:82)

Harmony $\sim$ -violation vector

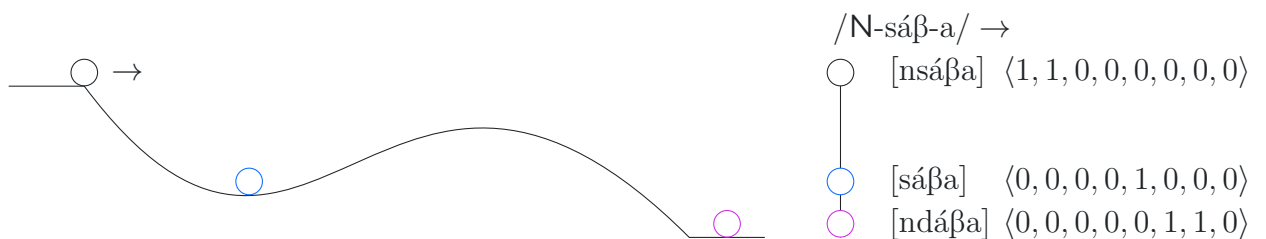
Not all attested mappings are global optima

- For a given constraint set
- Can introduce **conjoined constraints**¹¹ (Smolensky, 1993) or **weighted constraints**¹² (Farris-Trimble, 2008)
- But some mappings are instead **local optima**¹³
- McCarthy's (2007) potential energy analogy

¹¹ **conjoined constraints**

¹² **weighted constraints**

¹³ **local optima**



- If the grammar has to explore the candidate space step-by-step, it can get stuck in a local optimum

1.2 Harmonic Serialism works

Operations for Bukusu¹⁴

Feature changes $[\alpha F] \rightarrow [-\alpha F]$
 Deletion $/\alpha/ \rightarrow [\epsilon]$
 Epenthesis $/\epsilon/ \rightarrow [\alpha]$

Constraints same as before

¹⁴ For discussion of deletion as an operation see McCarthy (2008b, 2019) and Lin (2014) for epenthesis

(5) /N-tím-a/ → [n-dím-a]

/N-tím-a/	*[+nasal][−voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [Ntíma]	W 1					L		
b. [ʔtíma]			W 1			L		
c. [Nətíma]				W 1		L		
d. [tíma]					W 1	L		
☞ e. [Ndíma]						1		
f. [Ndíma]								
☞ g. [ndíma]								
☞ h. [ndíma]								

- [ndíma] is not a candidate in the first step
GEN cannot change [place] and [voice] simultaneously more on the second step in a future class
- Faithfulness constraints relative to input not UR
- Derivation **converges**¹⁵ once input is optimal

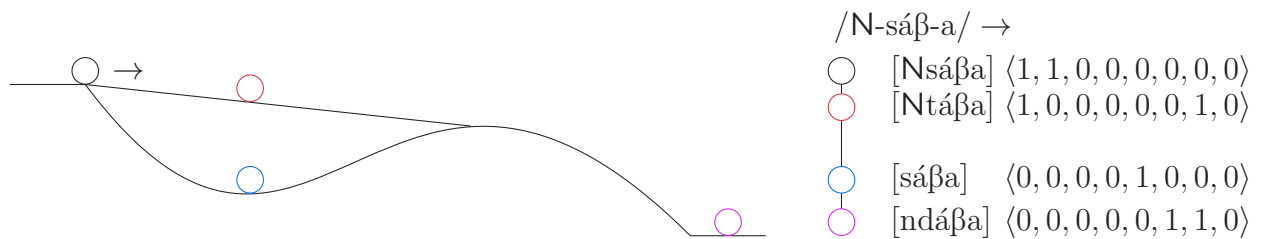
¹⁵ **converges**

(6) /N-sáβ-a/ → [sáβ-a]

/N-sáβ-a/	*[+nasal][−voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [Nsáβa]	W 1	W 1			L			
b. [ʔsáβa]			W 1		L			
c. [Nəsáβa]				W 1	L			
☞ d. [sáβa]					1			
e. [Náβa]					1			W 1
f. [Ntáβa]	W 1				L		W 1	
g. [Nzáβa]		W 1			L	W 1		
☞ h. [sáβa]								

- Deletion satisfies both phonotactic constraints
- [Ndáβa] is not a candidate in the first step

cannot change [continuant] and [voice] simultaneously
 every step picks optimal candidate; cannot reach [ndáβa]



- [sáβa] is local optimum in step 1, not global optimum
 HS is greedy: every step improves as much as possible

2 High tone in Jita (jit, Niger-Congo)

Tones spread one syllable to the right; let's ignore gliding and vowel coalescence

(7) Tone shift in Jita (Downing, 2014:103-106)

a.	/oku-βum-an-a/	[oku-βum-an-a]	'to hit each other'
b.	/oku-βón-an-a/	[oku-βon-án-a]	'to see each other'
c.	/oku-ljá munó/	[oku-lja múnó]	'to eat a lot'
d.	/oku-βón-a i:ɲopi/	[oku-βon-á i:ɲopi]	'to see a bird'
e.	/oku-mú-βum-a/	[oku-mu-βúm-a]	'to hit him/her'
f.	/oku-mú-βón-a/	[oku-mu-βón-a]	'to see him/her'
g.	/a-amá-gur-a/	[a-ama-gúr-a]	's/he bought'
h.	/a-amá-gósor-a/	[a-ama-gosór-a]	's/he visited'
i.	/a-amá-kú-gósor-er-a/	[a-ama-ku-gosór-er-a]	's/he visited you.sc'
j.	/a-tá-amá-gósor-a/	[a-ta-amá-gosór-a]	's/he did not visit'

- Bickmore (1999:130) posits a phonotactic constraint that requires high tone spans to contain exactly two TBUs, analogizing to FTBIN (see also McCollum and Kavitskaya, 2022)

Breteler (2017, 2018) derives binarity as footing

VVVV → V(VV)V → V(VV)V

2.1 Harmonic Serialism works

Operations for Ekegusii¹⁶

- Insert an association line (without creating a gap)
- Delete an association line (without creating a gap)
- Build a foot

Constraints

- LICENSE(H, F): AOVFE H tone not associated to a footed syllable¹⁷
- LICENSE(F, H): AOVFE foot that does dominate a H-toned syllable
- ALL-FEET-RIGHT: For every foot, assign as many violations as there are syllables to its right¹⁸
- ALIGN-R(H, F): AOVFE high tone span not right-aligned with a foot.
- DEP(link): AOVFE inserted association line¹⁹
- MAX(link): AOVFE deleted association line

¹⁶ For discussion of autosegmental operations see Kimper (2012), McCarthy et al. (2012), Gietz et al. (2023), and Lamont (2025) and for footing see Pruitt (2010, 2012)

¹⁷ Breteler (2018:19) after Zoll (1996)

¹⁸ McCarthy and Prince (1993:80)

¹⁹ Yip (2002:79)

(8) /ko-úmeran-a/ → [ko(úmé)rana], Step 1

H koumerana	LICENSE(H, F)	ALL-FEET-R	ALIGN-R(H, F)	DEP(link)	MAX(link)	LICENSE(F, H)
a. H koumerana	W 1	L	1			
b. H koumerana	W 1	L	1	W 1		
c. H koumerana	W 1	L	1		W 1	
d. H (kou)merana		W 3	L			
 e. H ko(ume)rana		2	1			
f. H koume(rana)	W 1	L	1			W 1

(9) /ko-úmeran-a/ → [ko(úmé)rana], Step 2 and convergence

	LICENSE(H, F)	ALL-FEET-R	ALIGN-R(H, F)	DEP(link)	MAX(link)	LICENSE(F, H)
H ko(ume)rana						
a. H ko(ume)rana		2	W 1	L		
☞ b. H / \ ko(ume)rana		2		1		
c. H ko(ume)(rana)		2	W 1	L		W 1
☞ d. H / \ ko(ume)rana		2				
e. H / \ ko(ume)rana		2		W 1		
f. H / \ ko(ume)(rana)		2				W 1

The SR is not the global optimum

(10) /ko-úmeran-a/ → [ko(úmé)rana] in parallel OT

	LICENSE(H, F)	ALL-FEET-R	ALIGN-R(H, F)	DEP(link)	MAX(link)	LICENSE(F, H)
H koumerana						
a. H koumerana	W 1	L	W 1	L		
☹ b. H / \ ko(ume)rana		2		1		
●* c. H / \ ko(ume)(rana)		2		1		

3 Local optionality

Variation as randomly varying ranking (Kimper, 2011)

French (fra, Indo-European)

(11) **Local optionality in French** (Kimper, 2011:424)

envie de te le demander

‘feel like asking you for it’

- [ãvi də tə lə dəmãde]
- [ãvi d_ tə lə dəmãde]
- [ãvi də t_ lə dəmãde]
- [ãvi də tə l_ dəmãde]
- [ãvi də tə lə d_ mãde]
- [ãvi d_ tə l_ dəmãde]
- [ãvi də t_ lə d_ mãde]

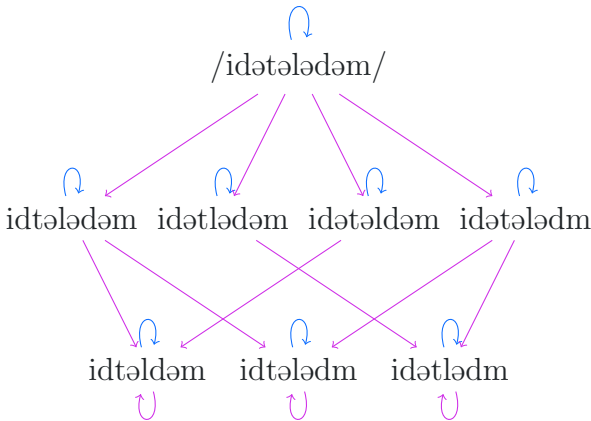
A gross oversimplification

- *CCC: AOVFE triconsonantal cluster
- *ə
- MAX

Operation: /ə/ → Ø (McCarthy, 2019)

*CCC ≫ {*ə, MAX}

(12) **All possible derivations;** MAX ≫ *ə vs. *ə ≫ MAX



(13)

/...idətələdəm.../	*CCC	*ə	MAX
a. [...idətələdəm...]		4	
b. [...idtələdəm...]		3	1
c. [...idətlədəm...]		3	1
d. [...idətəldəm...]		3	1
e. [...idətələdm...]		3	1

Not possible in parallel OT

(14)

	CCC	* _{CC}	MAX
/...idətələdəm.../			
a. [...idətələdəm...]		4	
 b. [...idtələdəm...]		3	1
 c. [...idətlədəm...]		3	1
 d. [...idətəldəm...]		3	1
 e. [...idətələdm...]		3	1
f. [...idtəldəm...]		2	2
g. [...idətlədm...]		2	2
h. [...idtəldəm...]		2	2

okay, but free choice is not a good model of variation

- EVAL cannot distinguish the tied candidates
- impossible to build in probabilistic preferences (McCarthy, 2008a:§6.2)

References

- Lee S. Bickmore. 1999. [High tone spread in Ekegusii revisited: An optimality theoretic analysis](#). *Lingua*, 109(2):109–153.
- Jeroen Breteler. 2017. [Deriving bounded tone with layered feet in Harmonic Serialism: The case of Saghala](#). *Glossa: a journal of general linguistics*, 2(1).
- Jeroen Breteler. 2018. [A foot-based typology of tonal reassociation](#). Ph.D. thesis, University of Amsterdam.
- Laura J. Downing. 2014. [Melodic verb tone patterns in Jita](#). *Africana Linguistica*, 20:101–119.
- Ashley W. Farris-Trimble. 2008. [Cumulative faithfulness effects in phonology](#). Ph.D. thesis, Indiana University.
- Frederick Gietz, Peter Jurgec, and Maida Percival. 2023. [Shift in Harmonic Serialism](#). *Journal of Linguistics*, 59(1):23–59.
- Wendell Kimper. 2011. [Locality and glob-ality in phonological variation](#). *Natural Language & Linguistic Theory*, 29(2):423–465.
- Wendell Kimper. 2012. [Harmony is myopic: Reply to Walker 2010](#). *Linguistic Inquiry*, 43(2):301–309.
- Andrew Lamont. 2025. [Shift is derived](#). *Journal of Linguistics*, 61(1):27–136.
- Kuo-Chiao Lin. 2014. [Gradual epenthesis: Echo vowels in Austronesian languages](#). *Oceanic Linguistics*, 53(2):443–466.
- John J. McCarthy. 2000. [Harmonic Serialism and Parallelism](#). In *Proceedings of NELS 30*, volume 2, pages 501–524, Amherst, MA. Graduate Linguistics Students Association.
- John J. McCarthy. 2006. [Restraint of analysis](#). In Eric Baković, Junko Itô, and John J. McCarthy, editors, *Wondering at the Natural Fecundity of Things: Essays in Honor of Alan Prince*, pages 195–219. Linguistics Research Center, Santa Cruz, CA.
- John J. McCarthy. 2007. [Restraint of analysis](#). In Sylvia Blaho, Patrik Bye, and Martin Krämer, editors, *Freedom of Analysis?*, volume 95 of *Studies in Generative Grammar*, pages 203–231. Mouton de Gruyter, Berlin and New York.
- John J. McCarthy. 2008a. [Doing Optimality Theory](#). Blackwell Publishing, Malden, MA.
- John J. McCarthy. 2008b. [The gradual path to cluster simplification](#). *Phonology*, 25(2):271–319.
- John J. McCarthy. 2019. [How to delete](#). In Amel Khalfaoui and Matthew Tucker, editors, *Perspectives on Arabic Linguistics XXX*, pages 7–32. John Benjamins, Amsterdam.
- John J. McCarthy, Kevin Mullin, and Brian W. Smith. 2012. [Implications of Harmonic Serialism for lexical tone association](#). In Bert Botma and Roland Noske, editors, *Phonological Explorations*, *Linguistische Arbeiten*, pages 265–298. De Gruyter, Berlin/Boston.
- John J. McCarthy and Alan Prince. 1993. [Generalized alignment](#). In Geert Booij and Jaap van Marle, editors, *Yearbook of Morphology 1993*, *Yearbook of Morphology*, pages 79–153. Springer Science+Business Media, Dordrecht.
- John J. McCarthy and Alan Prince. 1994. [The emergence of the unmarked: Optimality in prosodic morphology](#). In Mercè González, editor, *Proceedings of NELS 24*, volume 2, pages 333–379. Graduate Linguistics Students Association, Amherst, MA.
- John J. McCarthy and Alan Prince. 1995. [Faithfulness and reduplicative identity](#). In Jill Beckman, Suzanne Urbanczyk, and Laura Walsh Dickey, editors, *Papers in Optimality Theory*, pages 249–384. Graduate Linguistics Students Association, Amherst, MA.
- John J. McCarthy and Alan Prince. 1999. [Faithfulness and identity in prosodic morphology](#). In René Kager, Harry van der Hulst, and Wim Zonneveld, editors, *The Prosody-Morphology Interface*, pages 218–309. Cambridge University Press, Cambridge.
- Adam McCollum and Darya Kavitskaya. 2022. [On the status of non-iterativity in feature spreading](#). *Glossa: a journal of general linguistics*, 7(1).

- Nasiombe Mutonyi. 2000. *Aspects of Bukusu Morphology and Phonology*. Ph.D. thesis, The Ohio State University.
- Joe Pater. 1999. *Austronesian nasal substitution and other NÇ effects*. In René Kager, Harry van der Hulst, and Wim Zonneveld, editors, *The Prosody-Morphology Interface*, pages 310–343. Cambridge University Press, Cambridge.
- Alan Prince. 2002. *Arguing optimality*. In Angela Carpenter, Andries Coetzee, and Paul de Lacy, editors, *Papers in Optimality Theory II*, pages 269–304. Graduate Linguistics Students Association, Amherst, MA.
- Alan Prince and Paul Smolensky. 1993/2004. *Optimality Theory: Constraint Interaction in Generative Grammar*. Blackwell Publishing, Malden, MA.
- Kathryn Pruitt. 2010. *Serialism and locality in constraint-based metrical parsing*. *Phonology*, 27(3):481–526.
- Kathryn Pruitt. 2012. *Stress in Harmonic Serialism*. Ph.D. thesis, University of Massachusetts Amherst.
- Paul Smolensky. 1993. *Harmony, markedness, and phonological activity*. Paper presented at Rutgers Optimality Workshop 1.
- Maira Yip. 2002. *Tone*. Cambridge Textbooks in Linguistics. Cambridge University Press, Cambridge.
- Cheryl Zoll. 1996. *Parsing Below the Segment in a Constraint Based Framework*. Ph.D. thesis, University of California, Berkeley.
- Elizabeth Zsiga, Maria Gouskova, and One Tlale. 2006. *On the status of voiced stops in Tswana: Against *ND*. In *NELS 36: Proceedings of the Thirty-Sixth Annual Meeting of the North East Linguistic Society*, volume 2, pages 721–734, Amherst, MA. GLSA.